

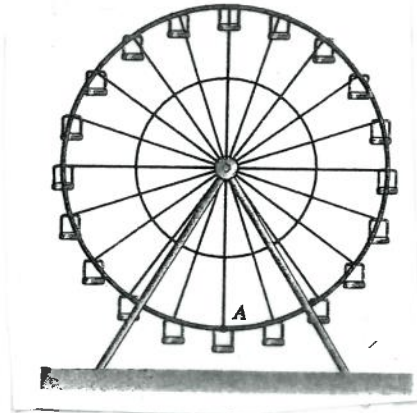
EMCH 310 Examination #1

ALL WORK MUST BE SHOWN TO RECEIVE ANY KIND OF CREDIT

NAME: Solution

Conceptual Questions

The Ferris wheel rotates. What is one of the directions of the acceleration at Point A?



- (a) $\rightarrow$
- (b) $\updownarrow$
- (c) $\downarrow$
- (d) $\leftarrow$
- (e) The acceleration is zero

Ball A is thrown up. Which of the following statements about the ball are true at the highest point in its path?

- a) The velocity and the acceleration are both 0
- b) The velocity is 0 but the acceleration is not 0
- c) The velocity is not 0, but the acceleration is 0
- d) Neither the velocity nor the acceleration are 0

In the absence of a net force, a moving object will

- a) Slow down and eventually stop
- b) Stop immediately
- c) Turn right
- d) Move with constant velocity
- e) Turn left

The position (meters) of an object as a function of time (seconds) is known to be $x(t) = 3t^3 - 4t^2 + 3t$. At 5 seconds, the acceleration (m/s^2) is most nearly (SHOW ALL WORK)

- a) 6.0
- b) 10.5
- c) 26.0
- d) 82.0

Take the derivative

$$x(t) = 3t^3 - 4t^2 + 3t$$

$$v(t) = 9t^2 - 8t + 3$$

$$a(t) = 18t - 8$$

$$a(5) = (18)(5) - 8$$

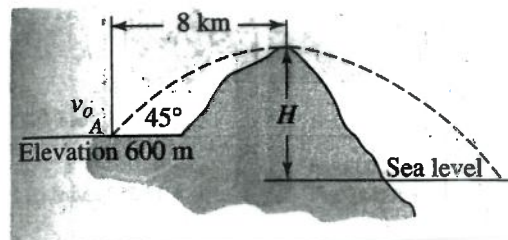
$$a(5) = 90 - 8$$

$$a(5) = 82 \text{ (m/s}^2 \text{ (1))}$$

SHOW ALL WORK.

A long-range artillery rifle located at Point A fires a shell at an angle of 45° with respect to the horizontal as illustrated. If the shell just barely clears the mountain peak at the top of its trajectory, the height "H" of the mountain peak above sea level is most nearly

- a) 7,998.0 m
- b) 4,600.0 m
- c) 4,000.0 m
- d) 12,600.0 m



At the top $V_y = 0$

Vertical Analysis

$$V_y = V_{0y} + at$$

$$0 = V_0 \sin 45^\circ - 9.8t$$

$$V_0 = \underline{13.86 \text{ (1)}}$$

Horizontal Analysis

$$x_2 - x_1 = V_{0x} t$$

$$V_{0x} = V_0 \cos 45^\circ$$

$$x_2 - x_1 = 8000 \text{ m (Given)}$$

$$8000 = 13.861 \cos 45^\circ t$$

$$t = \underline{28.6 \text{ s}}$$

$$y_2 - y_1 = V_{0y} t + \frac{1}{2} a_y t^2$$

$$y_2 - 600 = V_0 \sin 45^\circ t + \frac{1}{2} a_y t^2$$

$$y_2 = (13.86)(28.6)(0.707) + \frac{1}{2}(-9.81)(28.6)^2 + 600$$

$$H = y_2 = 8015 - 4012 + 600 = 4603 \text{ m} \approx 4600 \text{ m}$$

(B)

A motorist starts from rest at Point A on a circular entrance ramp when $t=0$ seconds and then increases the speed of her automobile at a constant rate and enters the highway at Point B.

Her speed continues to increase at the same rate until it reaches 35 m/s at Point C.

Determine

- The tangential components of the acceleration at Point B
- The distance at Point B
- The distance at Point C
- The speed at Point B

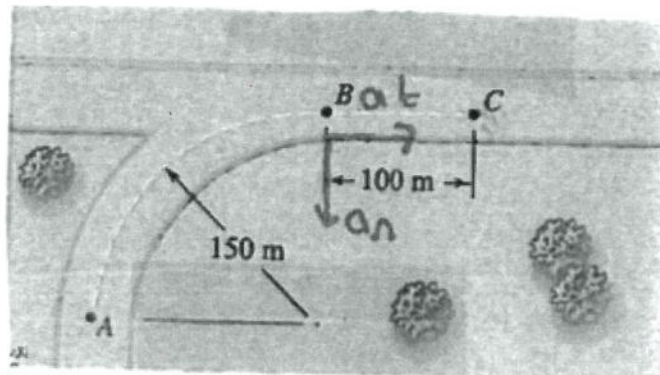
$V_0 = 0$ (motorist starts from rest)

$$a) V_1^2 = V_0^2 + 2a_t s$$

$$a_t = \frac{V_1^2 - V_0^2}{2s}$$

but $s = r\theta$
 $\theta = 90^\circ = \pi/2 \text{ radians}$
 $s = \frac{\pi(150)}{2} + 100$

$$s = 335.6 \text{ m}$$



$$a) a_t = \frac{(35)^2 - (0)^2}{2(335.6)} = 1.83 \frac{\text{m}}{\text{s}^2}$$

b) The distance at Point B $s_B = r\theta$ $\theta = 90^\circ = \frac{\pi}{2} \text{ radians}$

$$s_B = (150)\left(\frac{\pi}{2}\right) = 235.6 \text{ m}$$

c) The distance at Point C has already been shown in part a) $\approx 335.6 \text{ m}$

$$d) V_B^2 = V_0^2 + 2a_t s_B$$

$$V_B^2 = (0)^2 + (2)(1.83)(235.6)$$

$$V_B^2 = 862.3$$

$$V_B = 29.4 \frac{\text{m}}{\text{s}}$$

An athlete pulls handle A to the left with a constant acceleration.

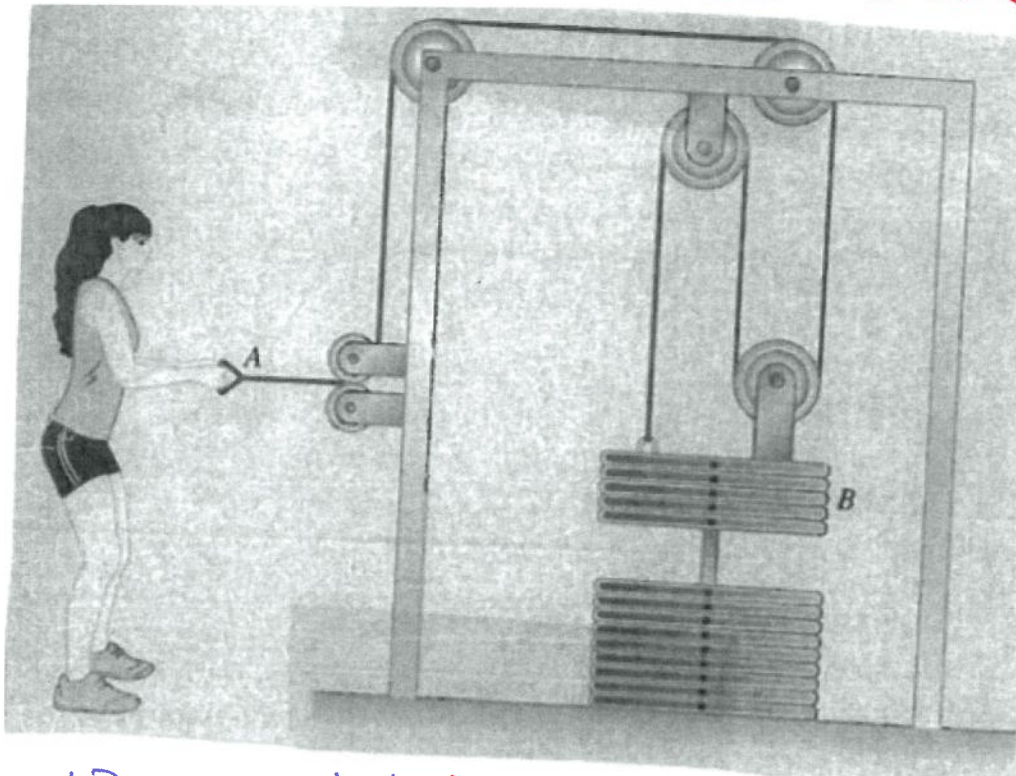
Weight B has been lifted 0.1 ft with a velocity of 5 ft/s. Assuming uniform accelerated motion, determine

- The acceleration of Weight B
- The acceleration of Handle A

UP = [-]

DOWN = [+]

x_A to the left
is positive



$$x_A + 3y_B = \text{constant} \quad (1)$$

$$v_A + 3v_B = 0 \quad (2)$$

$$a_A + 3a_B = 0 \quad (3)$$

$\Rightarrow$ The Dependent Equations

Uniform acceleration of weight stack B

$$v_B^2 = (v_B)_0^2 + 2a_B(y_B - y_{B0})$$

$$(-5)^2 = (0)^2 + 2a_B(-0.1 - 0)$$

$$25 = -0.2a_B$$

$$a) \quad a_B = -125 \text{ ft/s}^2 \uparrow \text{ UP}$$

$$b) \quad a_A = -3a_B = (-3)(-125) = 375 \text{ ft/s}^2 \text{ to the left } \leftarrow$$

The motion of a particle is defined by the relation $x = t^3 - 6t^2 + 9t + 5$, where x is expressed in feet and t in seconds. Determine

- The time when the velocity is zero.
- The position when $t = 5$ seconds.
- The acceleration when $t = 5$ seconds.
- The total distance traveled when $t = 5$ seconds.

$$a) v = \frac{dx}{dt} = 3t^2 - 12t + 9 = 0$$

$$t^2 - 4t + 3 = 0$$

$$(t-3)(t-1) = 0$$

$$t = 1 \text{ second and } t = 3 \text{ seconds}$$

$$\text{At } t = 1 \text{ second } x = (1)^3 - 6(1)^2 + 9(1) + 5 = 9 \text{ ft}$$

$$\text{At } t = 3 \text{ seconds } x = (3)^3 - 6(3)^2 + 9(3) + 5 = 5 \text{ ft}$$

$$b) \text{ At } t = 5 \text{ seconds } x = (5)^3 - 6(5)^2 + 9(5) + 5 = 25 \text{ ft}$$

$$c) a = \frac{dv}{dt} = 6t - 12$$

$$\text{at 5 seconds}$$

$$a = 6(5) - 12 = 18 \text{ ft/s}^2$$

$$d) 0 \leq t \leq 1 \quad x = |x_1 - x_0| = |9 - 5| = 4 \text{ ft}$$

$$1 < t < 3 \quad x = |x_3 - x_1| = |5 - 9| = |-4| = 4 \text{ ft}$$

$$3 < t \leq 5 \quad x = |x_5 - x_3| = |25 - 5| = 20 \text{ ft}$$

$$\text{Total Distance Traveled} = 4 + 4 + 20 = 28 \text{ ft}$$

The 100 kg block A and the 20 kg block B are released from rest. If the mass of the pulleys and the cord are neglected, determine the acceleration of Block A and the acceleration of Block B using the rectangular coordinate system.

The Dependent Equations

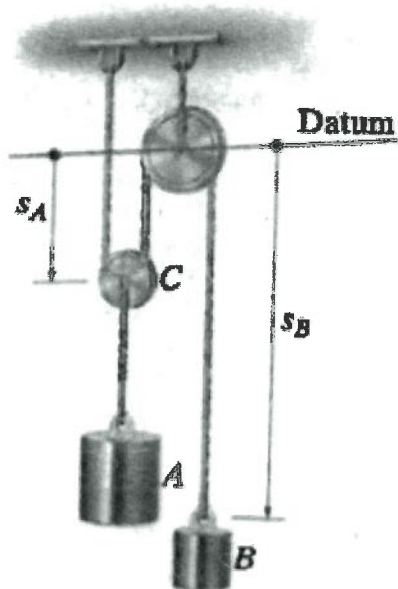
$$2s_A + s_B = \text{constant} \quad (1)$$

$$2v_A + v_B = 0 \quad (2)$$

$$2a_A + a_B = 0 \quad (3)$$

$$2a_A = -a_B$$

$$a_A = -\frac{1}{2}a_B \quad (4)$$



Block A Analysis ↓ is positive

$$\sum F_y = may$$

$$(100)(9.81) - 2T = 100a_A$$

$$981 - 2T = 100a_A \quad (5)$$

Block B Analysis ↓ is positive

$$\sum F_y = may$$

$$(20)(9.81) - T = 20a_B$$

$$196.2 - T = 20a_B \quad (6)$$

Substitute (4) into (5)

$$\begin{array}{rcl} 981 - 2T & = & -509B \quad (6) \\ 196.2 - T & = & 209B \quad (7) \end{array}$$

multiply (7) by 2

$$\begin{array}{rcl} 981 - 2T & = & -509B \\ -392.4 + 2T & = & -409B \\ \hline 588.6 & = & -909B \end{array}$$

$$a_B = -6.54 \text{ m/s}^2 \uparrow$$

$$a_A = 3.27 \text{ m/s}^2 \downarrow$$